

1. Mangooppers

- *Amritodus atkinsoni* (**largest**),
- *Idioscopus clypealis* (**smallest**),
- *Idioscopus niveosparsus* (**medium**)
- **Family:** Cicadellidae
- **Order:** Hemiptera
- **Host:** Mango
- **These insects are monophagous on mango and occur as regular pests.**

❖ Symptoms and Nature of Damage:

- During flowering, theoppers develop enormously in number, suck juice from the inflorescence and other tender plant parts reducing the vigour of the plant leading to reduction in fruit set and even premature fruit fall.
- The infestation also leads to development of sooty mould on the honeydew excreted by the insects.
- Egg laying also inflicts injury to the inflorescence. The infestation ranges from 25 to 50 per cent and in severe case it may lead to total loss of crop.
- During the remaining part of the year (off season), theseoppers occur in small numbers inside barks or on leaves of mango. High humidity in the air during flowering time encourages insect multiplication.
- Affected flowers wither and drop down.
- Presence of black sooty mould on floral and other tender plant parts.
- High humidity in orchards due to waterlogging, shading and overcrowding of trees favour the build-up of the pest.
- There are two peak generations of this insect during a year i.e. during February – April and June-August periods.

❖ Management

- Keeping orchard clean.
- Avoiding overcrowding and waterlogging.

- Proper pruning of the tree after harvesting to facilitate proper sunlight and air that minimises hopper population

❖ **Spray schedule is recommended as follows**

- At flower bud initiation, imidacloprid (0.036%) or carbaryl (0.1%)
- At emergence of inflorescence stalks and before flower opening (anthesis), dimethoate (0.03%) or thiamethoxam (0.025%) or imidacloprid (0.0036%).
- During anthesis and pollination, insecticides should not be sprayed.
- In case of very high population of hoppers at this stage, endosulfan 2ml/l is recommended
- When fruits are of pea size, monocrotophos (0.036WSC) or dimethoate (0.03%)
- Adding sulphur 3.5g/l to the insecticide based on need to check mites and sooty mould, *Capnodium mangiferum*
- Directing the spray first to stem/ trunk, then branches, twigs, leaves and finally inflorescence is a recommended method.

2. **Mango stem borer: *Batocera rufomaculata***

- **Family:** Cerambycidae
- **Order:** Coleoptera
- It is a **polyphagous** pest, infesting besides mango, apple, fig, mulberry, Eucalyptus, jack fruit, papaya etc.

❖ **Symptoms of Damage**

- Masses of grass and sap exuding from the bore holes.
- Leaves of damaged branches dry and fall.
- Branches collapse, tree succumbs in severe cases

❖ **Management**

- The affected portions with grubs and pupae removed and destroyed, if branches are affected.
- The bore holes traced and opened.
- A swab of cotton wool soaked in chloroform or petrol 5ml or carbon disulphide or kerosene oil or dichlorvos 10ml/l inserted into the hole and sealed with mud.
- Methyl parathion 1 ml/l poured into the hole or tablet of aluminium phosphide inserted into the hole to kill the grub.

- When burrows are superficial, extract the grubs with stiff hooked wire and paint bordeaux paste.

3. Mango fruit fly *Bactrocera dorsalis*

- **Family:** Tephritidae
- **Order:** Diptera
- It is one of the major pests of mango in India. It also infests guava, peach, citrus, ber, banana, papaya etc.

❖ Symptoms & Damage

- Semi ripe fruits with decayed spots.
- Dropping of fruits Damage to semi ripe fruits is caused by both maggot and the adult.
- The oviposition punctures made by the female serves as entry for fermenting organisms.
- Maggots feed on the pulp and convert the pulp into bad smelling discoloured semi liquid mass, unfit for use.
- The fruits develop brown rotten patches on them and fall to the ground eventually.

❖ Management

- Collection and destruction of fallen rotten fruits.
- Raking under the trees to expose the pupae.
- Mixing of carbaryl 10D in soil @ 50-100 g/tree.
- Install pheromone traps.
- Foliar spray with malathion (0.1%) + gur (2%) a month before harvesting the fruit crop repeated after 15 days.

❖ Post-Harvest Control (Heat treatment techniques):

- **Hot water treatment:** Submerging fruits in hot water at 43 to 46.7°C for 35- 90 min.
- **Double dip method:** Immersion of mango fruits in water at 40°C for 20 minutes, followed by 10 minutes at 46°C to get 100 per cent mortality of *Bactrocera dorsalis* eggs.

4. Mango mealybug: *Drosicha mangiferae*

➤ **Family:** Pseudococcidae

➤ **Order:** Hemiptera

❖ **Symptoms & Damage**

- Both nymph and adults suck sap from other tender plant parts thus reducing the plant vigour.
- Female lays eggs in clusters within ovisacs in soil under the trees 5-15 cm deep during April and May.
- The nymphs ascend the trees and settle on inflorescence.
- Nymphs and adults suck sap from inflorescence, fruit stalks, fruits *etc.* leading to flower drop, premature fruit drop *etc.*
- They also excrete honey dew on which sooty mould develops and the fruit development is hampered.

❖ **Management**

- Deep summer ploughing up to base of the tree trunks, after harvesting to expose eggs of mealy bugs.
- Dusting methyl parathion 2D or chlorpyrifos 5D around tree and incorporating into the soil.
- Spraying with acetamiprid (0.05%) or imidacloprid (0.0036%) or monocrotophos (0.036%), when severe mealybug infestation noticed on the twigs.
- Wrapping 25 cm wide, 400-gauge polythene sheet on the tree trunk 30 cm above ground level and pasting grease over it to prevent migration of freshly hatched first instar nymphs during winter (Nov-Dec) from soil to trees, one week before their emergence.
- Crawlers collecting beneath the polythene sheet may be scraped with a knife.

5. Other Pests

Common name	Scientific name	Family	Order
Mango Psylla	<i>Asylla cistellata</i>		
Mango Gall Midge	<i>Amradiplosis echinogalliperda</i> ,	Cecidomyiidae	Diptera

	<i>Procontarinia matteiana</i>		
Mango nut weevil or stone weevil	<i>Sternochetus mangiferae</i>	Curculionidae	Coleoptera
Mango fruit borer	<i>Deanolis albizonalis</i>	Pyralidae	Lepidoptera
Aphid	<i>Toxoptera odinae</i>	Aphididae	Hemiptera
Mango leaf webber	<i>Orthaga exvinacea</i>	Noctuidae	Lepidoptera
Termites	<i>Odontotermes obesus</i>	Termitidae	Isoptera

6. Pomegranate butterfly / Anar butterfly *Deudorix isocrates*

- **Family:** Lycaenidae
- **Order:** Lepidoptera
- It is the most important and destructive pest of pomegranate and distributed throughout the country, also infesting guava, annona, apple, ber, citrus, litchi, tamarind, wood apple, soap nut, etc.

❖ Symptoms & Damage

- The conspicuous symptoms are offensive smell and excreta of caterpillar at the entry hole.
- The affected fruits ultimately falling down.
- The fruit appears healthy but the caterpillar inside feeds on pulp and seeds just below the rind.
- It is only when the grown-up caterpillar comes out, a round hole is seen through which juices come out.

❖ Management

- Destruction of fallen infested fruits checks the spread.
- Removal of flowering weeds especially of Compositae family.
- Though expensive, bagging of fruits with polythene or paper bags or cloth bags soon after the fruit set prevents the pest attack.
- Initiate the spray schedule with the onset of flowering with any of following insecticides:
- Deltamethrin (0.0028%), cypermethrin (0.0075%), monocrotophos (0.036%), carbaryl (0.1%).

- Repeat the spray at 15-20-day interval.
- About 3 to 4 sprays are needed for effective control of the pest, as it continues to attack flowers (flowering in pomegranate remains for a longer time).

7. Guava mealy bug *Ferrisia virgata*

- **Family:** Pseudococcidae
- **Order:** Hemiptera
- It is otherwise called white tailed mealy bug /striped mealy bug.
- It is a widely distributed species in tropical and subtropical countries.
- It also infests bhindi, amaranthus, coccinia, colocasia, lab lab, tomato, brinjal, cashew, anona, guava, amla, grape, tobacco, pepper, cotton, **betelvine**

Pests of Grapevine

1. Grapevine Flea Beetle *Scelodonta strigicollis*

- **Family:** Chrysomelidae
- **Order:** Coleoptera
- It is the most destructive pest of grapevine all over India.

❖ Symptoms & Damage

- They feed on mature leaves cutting elongated holes on the leaf lamina like shot holes.
- The damage results in: Complete fed sprouting buds, Shot holes (rectangular cuttings) on mature leaves.
- Adult beetles hibernate during winter under tree bark and become active from March till November.
- Adults have characteristic habit of falling down and feigning death when disturbed.

❖ Management

- Adult beetles may be collected and killed.
- Removal of loose bark in rainy season after pruning to expose and eliminate eggs and adults found underneath.

- First spraying when buds swell in early morning or evening hours to kill beetles and second spray after 10 days with monocrotophos (0.036%) or carbaryl (0.1%) or imidachloprid (0.0072%) or cypermethrin (0.0075%)

2. Grapevine thrips: *Rhipiphorothrips cruentatus*

- **Family:** Thripidae
- **Order:** Thysanoptera
- Most destructive pest of grapevine India. It also feeds on rose, jasmine, cashew and other fruit trees.

❖ Symptoms & Damage

- Young nymphs on hatching feed on the under surface of leaves.
- Both the nymphs and adults' lacerate tender foliage and suck the oozing sap.
- The attacked leaves appear silvery initially and later turn brown and give withered appearance, curl up and drop off the plants.
- Severely affected vines do not bear fruits. If fruits are attacked, they develop corky layer on the fruits and turn brown.
- Infestation results in: Silvery patches on the affected leaves, Brown corky patches on fruits (scab)

❖ Management

- Removal of weeds in and around garden.
- Cutting of infested branches and burning.
- Spraying dimethoate (0.03%) or methyl demeton (0.025%) or thiamethoxam (0.25%)

3. Citrus butterfly *Papilio demoleus*, *Papilio polytes*, *Papilio helenus*

- **Family:** Papilionidae
- **Order:** Lepidoptera
- **Host:** It infests almost all citrus varieties though Malta (*Citrus sinensis*) is its preferred host.

- It can feed and breed on all varieties of cultivated or wild citrus and various other species of family Rutaceae.
- Besides citrus, it also attacks ber, wood apple, curry leaf.

❖ Symptoms and Damage

- Leaves fed up to midribs.
- Freshly hatched caterpillars are dark brown and soon develop irregular white markings on their body resembling bird's drop.
- The caterpillars feed voraciously on tender leaves right up to the mid ribs and defoliate the entire seedlings or the tree leaving behind the only midribs.

❖ Management

- In small orchards and nurseries with mild infestation, hand picking and destruction of various stages of the pest.
- Natural enemies enumerated below suppress the pest population
- **Egg parasitoids:** *Trichogramma evanescens*; *Telenomus* sp.
- **Larval parasitoid:** *Distatrix papilionis*; *Brachymeria* sp.
- **Pupal parasitoid:** *Pterolus* sp.
- Spraying of monocrotophos (0.036%) or cypermethrin (0.0075%) or deltamethrin (0.0028%)
- When the caterpillars are small. *B. t.* formulation HALT at 9 g/l is also recommended

4. Citrus leaf miner *Phyllocnistis citrella*

- **Family:** Gracillariidae
- **Order:** Lepidoptera
- It attacks all species of citrus but prefers sweet oranges. It also infests jasmine etc.

❖ Symptoms & Damage

- Characteristic silvery white zigzag galleries below the epidermis of tender leaves.
- Serious infestation causes retardation in growth. The infestation predisposes the leaves to canker growth.

- The tiny red, apodous larva mines in between the epidermal layers of the leaf in a zigzag manner and feeds on chlorophyll which results in distortion of the leaf lamina.
- The tunnel appears silvery white. New and tender leaves are preferred.
- Sometimes, the larva mines the outer layer of young green twigs.
- The full-grown caterpillar is slender, yellowish green with brownish mandibles.
- It is found inside the gallery formed in leaf tissue. Larval period is 15-30 days.
- Pupation takes place inside the leaf mine. Pupal period is 5-25 days.
- Total life cycle takes on an average 20-60 days depending on the climate.
- There are 9-13 overlapping generations in a year.

❖ **Management**

- Pruning of affected parts during winter and burning.
- Spraying of Neem cake solution 5% or neem oil 5% or monocrotophos (0.036%) or dimethoate (0.030%) or imidachloprid (0.0036%) twice at 10 days interval at every new flush time i.e., during June – July, Sep – Oct, Dec – Jan.

Common name	Scientific name	Family	Order
Citrus fruit sucking moths	Eudocima materna, Eudocima fullonica, Eudocima ancilla	Noctuidae	Lepidoptera
Citrus rust mite	Phyllocoptruta oleivora	Eriophyidae	Acarina
Citrus blackfly	Aleurocanthus wogluni	Aleurodidae	Hemiptera

PAPAYA

❖ **White fly, *Bemisia tabaci***

❖ **Symptoms of damage**

- Nymphs and adults suck the sap from under surface of the leaves
- Yellowing of leaves.

❖ **Management**

- Field sanitation